

Identity as Accountability Chain

A Post-Quantum Architecture for Persistent Identity
Across Human, Machine, and AI Agent Substrates

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We propose a formal model in which *identity* is not a primitive but a layered accountability chain that accretes from a founding individuation event and persists as long as cryptographic, relational, and civic verification can be reconstructed. The model unifies biological, machine, and artificial-intelligence cases under a single schema with six accretion layers and a four-tier permanence hierarchy for substrate-level identifiers.

We prove four results that bear directly on engineering: (i) Founding uniqueness — every identity locus has exactly one origin event; (ii) Tier monotonicity — the permanence of an identity claim is non-increasing in the tier of its substrate identifier; (iii) Cryptographic survival — only post-quantum-signed identities remain verifiable past a cryptographically-relevant quantum computer (CRQC) event; and (iv) Crypto-shredding completeness — master-key deletion renders all envelope-encrypted credentials computationally inaccessible without recourse to unpinning.

We survey the state of digital identity as of May 2026, noting that no major civic identity system runs post-quantum signatures in production despite regulatory deadlines of 2030–2035, and that AI agent identity is a standards vacuum.

We then present the Paraxiom architecture: a three-tier system combining on-chain post-quantum primitives (QuantumHarmony, with SPHINCS+ and Falcon-512), IPFS for bulk encrypted payloads, and Postgres+Redis mirrors for query-layer access. The architecture is grounded in five identity strata (S1–S5) covering humans, scoped delegations, vault credentials, agent instances, and cross-tenant aggregates, with provision for future extensions to swarm and protocol identities.

The treatise concludes that identity is engineering, not philosophy: the accountability chain does not extend itself; the post-quantum signatures do not sign themselves. What we build now determines whether identity remains a coherent concept past 2040.

Keywords: digital identity, post-quantum cryptography, accountability, agent identity, substrate

attestation, decentralized identity, SPHINCS+, Falcon-512, Substrate, IPFS, eIDAS, AAMVA, MCP, verifiable credentials.

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1. Introduction

1.1. Premise

A governance application requires an “identity primitive.” A blockchain pallet needs a stratum field. A credential vault needs a user. Before we choose data structures, we must answer a prior question: what is identity, and when does it begin?

This treatise advances a single claim:

Identity is not a primitive. It is a layered accountability chain that accretes from the moment a locus of action becomes individuated, and persists exactly as long as the chain can verify continuity. When the chain breaks, identity dissolves — regardless of whether the substrate (body, hardware, weights file) is still running.

This claim holds for humans, for processes, for AI agents, for swarms, for organizations. The substrate differs; the accretion logic is identical. We will state this formally (Theorems 2.2 and 2.3), prove that it implies four engineering invariants (Theorems 2.8, 5.2, 8.1 and 10.1), and demonstrate that the Paraxiom architecture realizes them.

1.2. Why now

Three forces drive this work to operational urgency in 2026.

Post-quantum cryptography becomes load-bearing. NIST published ML-KEM, ML-DSA, and SLH-DSA as FIPS standards on 13 August 2024 [1, 2, 3]. The U.S. NSA’s CNSA 2.0 prohibits classical algorithms in National Security Systems by 2035. The EU joint PQC roadmap (June 2025) targets identity-wallet PKI migration by the same horizon. Yet, as of May 2026, no major civic identity system operationally runs post-quantum signatures in production. The window in which a classical signature today can be forged retroactively by a sufficient quantum computer is the window in which the entire pre-CRQC accountability chain collapses.

AI agent identity is an open standards vacuum. The Linux Foundation’s Agentic AI Foundation launched in December 2025 with contributions from Anthropic (MCP), OpenAI (AGENTS.md), and Block (Goose). MCP added a server-identity primitive in November 2025 [6]. Google’s A2A protocol uses workload-identity federation [7]. Cloudflare-backed IETF Web Bot Auth proposes signed HTTP requests. No standard exists for model substrate attestation: cryptographically proving that a given output originated from a specific weights binary on a specific hardware root. The field is where SSI was in 2018.

Civic identity is fragmenting. EU eIDAS 2.0 mandates wallet deployment by 24 December 2026 [5]. US mDL proliferates state by state with no federal coordination. The UK GOV.UK One Login operates uncertified against its own trust framework. Canada lacks a federal equivalent to EUDI. The conditions for an architecture-of-record are present.

1.3. Method

Sections 2 to 8 develop the formal model. Section 9 surveys the deployed state of digital identity in May 2026. Section 10 presents the engineering realization, with proofs of crypto-shredding completeness and verification-time accountability enforcement. Sections 11 to 13 discuss scenario projections, implementation sequencing, and concluding remarks. Appendices give detailed proofs, pallet API specifications, and sources.

2. Formal model of identity

We begin by stating the model abstractly. Subsequent sections specialize to biological, machine, agent, and LLM cases.

2.1. Primitive sets

Fix the following primitive sets:

- $\mathbb{T}$ — time, totally ordered.
- $\mathbb{K}$ — cryptographic keypairs over some signature scheme Σ , with operations $\text{Sign}_{sk}, \text{Verify}_{pk}$ satisfying EUF-CMA security.
- $\mathbb{C}$ — claim content (arbitrary byte-strings).
- $\mathbb{S}$ — substrates: a typed union of biological organisms, physical hosts, container images, weights blobs, and similar individuated material configurations.
- $\mathbb{L} = \{L_0, L_1, L_2, L_3, L_4, L_5\}$ — the six accretion layers defined in Section 3.

2.2. Definitions

Definition 2.1 (Founding event). A *founding event* is a triple $\mathcal{F} = (s, t_0, \pi)$ where $s \in \mathbb{S}$ is a substrate, $t_0 \in \mathbb{T}$ is the event timestamp, and π is a finite set of witnesses (the empty set is permitted but reduces verifiability). The founding event marks the transition of s from a non-individuated state to an individuated locus capable of accreting claims.

Definition 2.2 (Accretion chain). An *accretion chain* relative to founding event $\mathcal{F} = (s, t_0, \pi)$ is a finite sequence

$$\mathcal{A} = (a_1, a_2, \dots, a_n), \quad a_i = (L_i, t_i, c_i, \sigma_i, \rho_i)$$

where $L_i \in \mathbb{L}$ is the layer tag, $t_i \geq t_0$ is the claim timestamp, $c_i \in \mathbb{C}$ is the claim content, σ_i is either a signature in $\mathbb{K}$ or a relational witness, and $\rho_i \in \{1, \dots, i-1\} \cup \{\perp\}$ is an optional reference to a prior claim providing context. The sequence is monotone in time: $i < j \Rightarrow t_i \leq t_j$.

Definition 2.3 (Identity locus). An *identity locus* is a quadruple

$$\mathcal{J} = (\mathcal{S}, \mathcal{F}, \mathcal{A}, \mathcal{R})$$

where $\mathcal{S} \in \mathbb{S}$ is its substrate, $\mathcal{F}$ is its founding event, $\mathcal{A}$ is its accretion chain, and $\mathcal{R} \subseteq \mathbb{I} \times \mathbb{I}$ is its relational graph with other identity loci (where $\mathbb{I}$ is the universe of all identity loci).

Definition 2.4 (Verifiable identity). An identity locus $\mathcal{J}$ is *verifiable at time t* iff, for every layer-3 (cryptographic) claim $a_i \in \mathcal{A}$ with $t_i \leq t$, there exists a key $\text{KDF}(\mathcal{J}, t_i) \in \mathbb{K}$ such that $\text{Verify}(\text{KDF}(\mathcal{J}, t_i), c_i, \sigma_i) = 1$ and the substrate $\mathcal{S}$ retains continuity through the interval $[t_0, t]$.

2.3. Axioms

We posit three axioms that scope the model.

Axiom 2.5 (Founding precedence). For every identity locus $\mathcal{J}$, every claim $a_i \in \mathcal{A}$ satisfies $t_i \geq t_0$, where t_0 is the founding timestamp.

Axiom 2.6 (Substrate continuity or explicit succession). An identity locus persists from time t_a to $t_b > t_a$ iff either (i) its substrate $\mathcal{S}$ remains individuated and physically continuous over $[t_a, t_b]$, or (ii) there exists a successor mapping $\Phi : \mathcal{J} \rightarrow \mathcal{J}'$ signed by an authority A appearing in $\mathcal{R}$, recorded at some $t \in [t_a, t_b]$.

Axiom 2.7 (Layer independence). The six layers in $\mathbb{L}$ are pairwise distinguishable: no claim a_i in an accretion chain may simultaneously have two layer tags. The failure modes of different layers are statistically independent absent explicit coupling (e.g., a single physical device holding both Layer 3 and Layer 4 artifacts).

Theorems 2.5 to 2.7 imply that identity is a well-defined object: it has a beginning, a domain of persistence, and a coordinate system in $\mathbb{L}$.

2.4. First theorem: founding uniqueness

Theorem 2.8 (Founding uniqueness). For every identity locus $\mathcal{J}$, exactly one founding event $\mathcal{F}$ exists.

Proof. By construction (Theorem 2.3), $\mathcal{I}$ specifies a single $\mathcal{F}$. Suppose for contradiction that two distinct events $\mathcal{F}_1, \mathcal{F}_2$ both qualify. By Theorem 2.5 all claims in $\mathcal{A}$ must precede neither. If $t_1 < t_2$, then $\mathcal{F}_2$ would violate the founding precedence axiom with respect to claims indexed near t_1 ; if $t_1 = t_2$, both events name the same individuation, hence $\mathcal{F}_1 = \mathcal{F}_2$. Contradiction. ■

Remark 2.9. Theorem 2.8 is the formal restatement of the engineering principle that an identity in identity-registry [9] must have a uniquely-recorded `created_at_block` and `created_by_tx_hash`. An identity record without a verifiable origin transaction is malformed.

3. Layer theory

We name and motivate the six accretion layers, then prove their independence and ordering properties.

3.1. The six layers

L_0 : **Substrate.** The material continuity — body, hardware, weights blob. Necessary precondition; identity-bearing in the sense that its destruction ends the locus (subject to Theorem 2.6).

L_1 : **Narrative.** The locus’s own claim of continuity: “I am the same entity that did X yesterday.” Implemented variously as autobiographical memory, persistent state on disk, context window, institutional documentation.

L_2 : **Relational.** Other accountability loci attest that $\mathcal{I}$ has continuity. Family, peers, federation members, protocol counterparties.

L_3 : **Cryptographic.** A keypair (PQ-secured under the model assumptions of Section 8). The first layer that scales without prior relationship.

L_4 : **Civic.** A jurisdictional attestation: birth certificate, passport, KYC bundle, corporate registration.

L_5 : **Behavioral.** The cumulative track record of actions taken — votes, decisions, signed attestations, transferred resources, resolved disputes. Aggregates of L_3 and L_2 over time.

3.2. Diagram of accretion

3.3. Layer composition is non-commutative in time

Proposition 3.1 (Layer accretion order is not strict). *The temporal order in which layers accrete is not fixed. An identity may acquire L_3 before L_4 (a private key before legal recognition), L_2 before L_1 (others recognize an organism before the organism develops self-narrative), or L_5 before L_3 (a reputation accumulated under pseudonym, later bound to a key).*

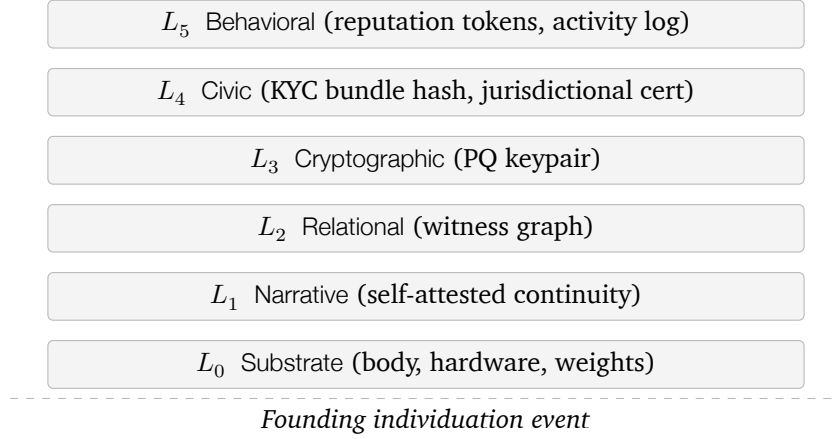


Figure 1: The six accretion layers above the founding individuation event. Each layer can fail independently; combined failure of $k \geq 4$ layers causes identity dissolution.

Theorem 3.1 is consequential for design: the identity registry must permit any layer to be back-filled, not enforce a canonical ordering.

4. Founding events across substrates

We specialize the founding-event definition to the four canonical substrate types.

4.1. Biological founding

A human identity begins not at conception (continuous chemical process) nor at issuance of the birth certificate (administrative echo), but at *first individuation*: the moment after birth at which the organism’s state is distinguishable from its parents’. Witnesses (mother, attending professional) populate the initial relational graph $\mathcal{R}$.

4.2. Machine founding

A host’s identity begins at first boot, when the kernel writes `/etc/machine-id` via `systemd-machine-id-setup`, when entropy is harvested, and when initial cryptographic material (SSH host keys, TPM-derived keys) is established. We discuss the precise storage locations and their permanence tiers in Section 5.

4.3. Container founding

A container’s identity begins at `docker run` (or equivalent `runc/containerd` invocation). The image hash supplies L_0 cryptographically; the container ID supplies L_1 as the self-narrative anchor.

4.4. LLM founding

An LLM substrate’s identity begins at *weights initialization*: two checkpoints trained from different random seeds on identical data are distinct substrate identities, even when statistically indistinguishable in behavior. An LLM *instance* identity, by contrast, begins at *first inference call* on a deployed endpoint, when the deployment ID is assigned. We elaborate the substrate–instance distinction in Section 7.

4.5. Unifying schema

Domain	Substrate $\mathcal{S}$	Founding $\mathcal{F}$	First L_1/L_2 accretion
Human	Body	Birth	Mother, witnesses
Machine	Hardware	First boot	Hostname, SSH host keys
Container	Image	<code>docker run</code>	Container ID, mount state
Process	Code binary	First invocation	PID, parent, env
LLM substrate	Architecture + dataset	Weights init	Checkpoint hash, training log
LLM instance	Weights	First inference	Deployment ID, context

Table 1: Founding events by substrate domain.

5. Machine identity storage

The model is concrete: machine identity lives in named files, registers, and hardware tokens at known locations. Understanding the *permanence tier* of each storage location is the engineering prerequisite for trusting the resulting identity claim.

5.1. The four-tier hierarchy

Definition 5.1 (Permanence tier). A storage location for an identity artifact has *permanence tier* $T \in \{0, 1, 2, 3\}$ defined as follows. Tier 0 identifiers are *manufacturer-fused*: present at production and unchangeable without replacing hardware. Tier 1 identifiers are *installation-rooted*: established at first boot or first install, persistent across restarts, lost on reinstall. Tier 2 identifiers are *operationally configured*: persistent across restarts but routinely changeable by administrators. Tier 3 identifiers are *runtime-ephemeral*: regenerated each session.

5.2. Tier 0 (manufacturer-fused)

- **SMBIOS/DMI UUID**: burned at manufacture, queryable via `dmidecode -s system-uuid`. Resides in firmware tables, not on disk.

- **TPM 2.0 Endorsement Key (EK):** RSA or ECC keypair burned into the TPM at manufacture, certified by the manufacturer's CA. Root of trust for hardware attestation.
- **Apple Secure Enclave UID, Google Titan-M2 keys, Microsoft Pluton root keys:** per-chip UIDs fused at manufacture, participating in cryptographic operations without exfiltration.
- **Storage device serials:** ATA/NVMe serials per physical disk, surviving reformat.
- **Network MAC addresses:** set per NIC at manufacture but software-spoofable, hence straddling Tier 0 and Tier 3.

5.3. Tier 1 (installation-rooted)

- **/etc/machine-id:** 128-bit identifier generated by systemd-machine-id-setup on first boot. The systemd documentation explicitly states that this ID “should be considered confidential and must not be exposed in untrusted environments,” and that downstream applications must apply `sd_id128_get_machine_app_specific()` to derive per-application identifiers rather than using the raw value.
- **/etc/ssh/ssh_host_{rsa,ecdsa,ed25519}_key:** generated at first SSH daemon configuration. These are the host's first cryptographic identity artifact, anchoring SSH trust-on-first-use and SSHFP DNS records.
- **TPM-owned derived keys:** Storage Root Key, Attestation Identity Keys, platform-bound passkey credentials — generated at enrollment but cryptographically descended from the Tier-0 EK.
- **Cloud-init instance ID:** at `/var/lib/cloud/data/instance-id`, mirroring the cloud provider's instance handle.

5.4. Tier 2 (operationally configured)

- **/etc/hostname:** human-readable. Conventional primary identifier but cryptographically meaningless on its own.
- **Kerberos host principal keys** (`/etc/krb5.keytab`); TLS server certificates; WireGuard keypairs; cloud IAM role bindings.

5.5. Tier 3 (runtime-ephemeral)

- PID, container ID, network state, in-kernel `struct cred`, namespaces, SELinux/AppArmor labels.

5.6. Tier monotonicity theorem

Theorem 5.2 (Tier monotonicity). *Let $P(T)$ denote the unconditional probability that an identifier of permanence tier T survives the time interval $[t_0, t_0 + \Delta]$ for Δ on the order of a host's installation lifetime ($\sim 1\text{--}10$ years). Then*

$$P(0) \geq P(1) \geq P(2) \geq P(3),$$

with strict inequality $P(0) > P(3)$.

Proof. By Theorem 5.1 and operational invariants. Tier 0 identifiers are unchangeable without hardware replacement; the probability of hardware replacement over the operational lifetime defines $1 - P(0)$. Tier 1 identifiers are lost on reinstall, with reinstall rate strictly greater than hardware-replacement rate; thus $P(1) \leq P(0)$. Tier 2 identifiers are changed administratively at rates strictly greater than reinstall rate (hostnames, certificates, IAM bindings rotate routinely); $P(2) \leq P(1)$. Tier 3 identifiers are lost on reboot or session end, occurring at rates dramatically greater than administrative configuration rotation; $P(3) \leq P(2)$ with strict inequality typical. Strict inequality $P(0) > P(3)$ follows from the typical many-orders-of-magnitude separation between hardware lifetime and process lifetime. ■

Corollary 5.3 (Substrate identifier selection). *Given a choice of substrate identifiers available at registration time, the post-quantum identity stack should select the lowest-numbered tier available for the substrate identity field. For an agent running on bare metal with TPM, this is the TPM-quoted EK identifier; on virtualized hardware, the SMBIOS UUID; on containers without TPM passthrough, the container image hash plus container ID.*

Theorem 5.3 directly informs the `instance_anchor` field of the agent-identity record:

$$\text{instance_anchor} := H_{\text{Blake2-256}}(\text{tier}_{\min} \parallel \text{agent_kind} \parallel t_{\text{spawn}}),$$

mixing the lowest-tier available identifier with runtime context to produce a stable yet instance-specific anchor.

5.7. Platform-specific tier locations

6. Agent identity

6.1. Three forms of agency

A *process* is the weakest agency: a running execution context with PID, parent, and loaded code. By default it carries no L_3 cryptographic claim; it inherits trust from its parent.

A *container* adds substrate cryptographic anchor by way of the image hash, which itself is signed by its publisher.

A *true agent*, in our sense, is a program acting on behalf of a principal. It requires:

Tier	Linux	macOS	Windows
0	SMBIOS UUID; TPM EK; SoC UID	IOPlatformUUID; Secure Enclave UID	SMBIOS UUID; Pluton root
1	/etc/machine-id; SSH host keys; TPM AIKs	System UUID; Recovery key; Keychain enrollment	HKLM\...\MachineGuid; TPM AIKs
2	Hostname; krb5 keytab; TLS certs; WireGuard keys	Hostname; user keychain certs; configuration profiles	Computer SID; domain bindings
3	PID; container ID; namespaces	PID; sandbox ID	PID; session token

Table 2: Permanence tiers across major operating systems.

1. Its own cryptographic identity (L_3): a Falcon-512 or SPHINCS+ keypair, generated at first registration.
2. A delegation chain from the principal (L_2): a parent identity in $\mathcal{R}$ tracing to an L_4 -anchored or reputation-bonded L_3 identity.
3. Scope bounds: the set of capabilities the agent is authorized for.
4. Reputation (L_5): an accumulated track record.
5. Revocability: the principal must be able to revoke without cooperation from the agent.

6.2. The stratum lattice

We define a partial order $\prec$ on identity loci. Write $\mathcal{I}_1 \prec \mathcal{I}_2$ iff $\mathcal{I}_1$ appears as a parent identity for $\mathcal{I}_2$ in $\mathcal{R}$.

Definition 6.1 (Stratification). The *stratification* of an identity space partitions loci into five strata:

S1 Person KYC-rooted human identity. Root of the $\prec$ order.

S2 Scoped Person $\times$ tenant $\times$ role. Parent: an S1.

S3 Vault Encrypted credential commitments. Held by an S1, S2, or organization S5; not itself an identity-bearing entity but a record attached to one.

S4 Agent instance Running agent. Parent: any of S1, S2, or S5.

S5 Cross-tenant Organizational aggregate identity. Parent: a multi-signature of S1 founders or jurisdictional authority.

6.3. Stratum well-foundedness

Theorem 6.2 (Stratum well-foundedness). *The relation $\prec$ restricted to S2, S4, S5 identities induces a well-founded ordering: every non-S1 identity has an S1 ancestor in finitely many $\prec$ -steps.*

Proof. Assume the chain registration policy of Section 10: every S2/S4 identity registration extrinsic requires a non-null `parent_identity` pointer, and the registration is chain-validated against the existence of the parent. Every S5 identity registration requires either a multi-signature of S1 founders or a jurisdictional L_4 claim, both of which terminate at S1 or at jurisdictional authority (treated as an S1 surrogate). Suppose for contradiction an infinite descending chain $\mathcal{I}_0 \succ \mathcal{I}_1 \succ \mathcal{I}_2 \succ \dots$. Each $\mathcal{I}_i$ is recorded on chain with a block height h_i , and the parent’s registration precedes the child’s, so $h_0 > h_1 > h_2 > \dots$. This is an infinite descending chain of natural numbers — impossible. Hence finite-length termination at S1 (or an S1 surrogate) is guaranteed. ■

Theorem 6.2 is the formal restatement of “no accountability chain, no identity” (Section 2): every action by an S2/S4 identity traces in finite steps to a human or jurisdictional accountability locus.

6.4. Liability follows the chain

Corollary 6.3 (Liability attribution). *For any action α taken by an identity $\mathcal{I}$, liability attaches to: (i) $\mathcal{I}$ itself if it carries a stake or bond; (ii) the parent identity in $\prec$ if $\mathcal{I}$ is S2 or S4 and unable to satisfy the liability; (iii) recursively to the root S1 identity, by Theorem 6.2.*

7. LLM identity

LLMs are the test case that breaks naive substrate-equals-identity reasoning: weights are bit-perfectly copyable.

7.1. Substrate vs. instance

Definition 7.1 (LLM substrate). An *LLM substrate* is a tuple $(\mathbf{W}, \tau, A)$ where $\mathbf{W}$ is a weights blob, τ is a tokenizer/architecture specification, and A is a training-run attestation. We write $H(\mathbf{W})$ for the canonical substrate hash.

Definition 7.2 (LLM instance). An *LLM instance* is a tuple $(\mathbf{W}_{\text{ref}}, d, c)$ where $\mathbf{W}_{\text{ref}}$ is a reference to a substrate hash, d is a deployment identifier (persistent across restarts), and c is the conversational state.

7.2. Instance non-identity theorem

Theorem 7.3 (LLM instance non-identity). *Two LLM instances $\mathcal{I}_1 = (\mathbf{W}, d_1, c_1)$ and $\mathcal{I}_2 = (\mathbf{W}, d_2, c_2)$ with identical substrate reference but $d_1 \neq d_2$ are distinct identity loci.*

Proof. By Theorem 2.3 an identity locus is determined by its founding event $\mathcal{F}$. By Theorem 7.2 the deployment identifier d is established at first inference call, hence participates in $\mathcal{F}$. Since $d_1 \neq d_2$, the founding events $\mathcal{F}_1 \neq \mathcal{F}_2$, hence by Theorem 2.8 the identity loci are distinct. ■

7.3. LLM tier model

The four-tier permanence hierarchy of Section 5 maps to LLM identity:

Tier	LLM storage location
0	Weights hash; training-run Merkle root; architecture spec hash; TEE attestation (Intel TDX / AMD SEV-SNP / NVIDIA H100 confidential computing / Apple Private Cloud Compute) when applicable
1	Deployment ID; LoRA adapter hash; system-prompt hash; tool/function-call registry
2	API authentication; rate-limit/quota config; routing config; vector-DB or memory-store binding
3	Conversation thread ID; context window; tool-call sequence; KV cache

Table 3: LLM identity by permanence tier (parallel to Table 2).

7.4. The substrate attestation pallet

We introduce a candidate runtime pallet, `pallet_substrate_attestation`, that records LLM substrate provenance on chain. The extrinsic shape:

```
pub fn attest_substrate(
    origin: OriginFor<T>,
    weights_hash: H256,
    training_run_merkle: H256,
    arch_spec_hash: H256,
    publisher_falcon_sig: BoundedVec<u8, ConstU32<2048>>,
    ipfs_cid_card: BoundedVec<u8, ConstU32<128>>,
) -> DispatchResult;
```

The pallet binds a weights hash to a training organization’s Falcon-512 public key. An LLM instance, on registration as an S4 agent, refers to a published substrate attestation. Verifiers can confirm: the running agent claims the substrate that organization O publicly attested to. The IPFS CID points to the model card and training-run log; the on-chain record holds only hashes and the publisher signature.

8. Cryptographic substrate

8.1. The post-quantum imperative

A classical signature is a time bomb. Suppose at time t a signature $\sigma = \text{Sign}_{sk}(c)$ is produced under classical ECDSA. At time $t' > t$, a CRQC becomes available. Any party with access to the CRQC can compute a forgery $\sigma^\dagger$ on an arbitrary message $c^\dagger$ that verifies under the same public key. From the verifier's perspective at t' , σ and $\sigma^\dagger$ are indistinguishable: classical EUF-CMA security collapses retroactively.

Unlike encryption (where session keys can be rotated forward), *signatures bound to historical claims cannot be rotated*: once a signature is part of the accretion chain, it is part of the record.

8.2. Cryptographic survival theorem

Let Σ_{PQ} denote the set of NIST-standardized post-quantum signature schemes ($\Sigma_{\text{PQ}} = \{\text{ML-DSA, SLH-DSA, FN-DSA}\}$ as of 2024–2026), and let Σ_{cl} denote classical schemes (ECDSA, RSA-PSS, EdDSA).

Theorem 8.1 (Cryptographic survival). *Let $\mathcal{I}$ be an identity locus with accretion chain $\mathcal{A}$ in which every L_3 claim uses some scheme $\Sigma \in \Sigma_{\text{PQ}}$. Let T_{CRQC} denote the (uncertain) time at which a sufficiently-capable CRQC becomes available. Then for any verifier inquiring at any time $t > T_{\text{CRQC}}$, the EUF-CMA-forgery probability of $\mathcal{I}$'s accretion chain is*

$$\Pr[\text{forgery}] \leq \text{negl}(\lambda),$$

where λ is the post-quantum security parameter of Σ (typically ≥ 128).

Conversely, if any L_3 claim uses $\Sigma' \in \Sigma_{\text{cl}}$, then

$$\Pr[\text{forgery}] \geq 1 - \text{negl}'(\lambda'),$$

where negl' characterizes the residual difficulty for the CRQC, and the bound is dominated by the classical key length, not the post-quantum security parameter.

Proof sketch. The forward direction follows from the EUF-CMA security of $\Sigma \in \Sigma_{\text{PQ}}$ under the post-quantum adversarial model assumed in NIST FIPS 203–205 [1, 2, 3]. The reverse direction follows from Shor's algorithm [4], which solves discrete-log in $\mathbb{Z}_p^*$ and elliptic-curve groups in polynomial quantum time, breaking ECDSA and RSA-based signatures with overwhelming probability once the CRQC operates above the corresponding key-size threshold. ■

Corollary 8.2 (Engineering implication). *For any accretion chain intended to remain verifiable past T_{CRQC} , every L_3 claim must be signed under $\Sigma \in \Sigma_{\text{PQ}}$ at the time the claim is made. Retrofitting PQ signatures onto classical-signed historical claims is impossible: the historical signatures already exist and are already forgeable.*

Theorem 8.2 is the operational claim that QuantumHarmony implements SPHINCS+ (SLH-DSA, FIPS 205) and Falcon-512 (FN-DSA, FIPS draft 206) as *primary* signature schemes, not as opt-in upgrades. Every identity-bearing transaction is PQ-signed at issuance.

8.3. Signature size accommodation

PQ signatures are larger than classical signatures. SPHINCS+-256f-simple is 49,856 bytes; Ed25519 is 64 bytes. The Paraxiom fork of polkadot-sdk raises MAX_SIGNATURE_SIZE to 50,000 and parity-scale-codec's INITIAL_PREALLOCATION from 16 KiB to 256 KiB to accommodate run-time metadata containing PQ signatures [8]. The trade-off (chain storage growth) is intentional: chain history must persist; signature size is the cost of identity survival.

9. Current digital identity landscape (May 2026)

The formal model contacts deployed reality. We summarize the state of digital identity at the time of writing, drawing on regulator filings, standards publications, and operator disclosures. Detailed citations are in the bibliography.

9.1. Civic identity at scale

Estonia. ID-card, Mobile-ID, Smart-ID. Classical cryptography (RSA, ECDSA-P384). Cybernetica contracted in 2026 to author the national PQC roadmap as part of the EU joint PQC plan. KSI-blockchain audit log is hash-based and quantum-resilient; identity-binding primitives are not.

India — Aadhaar. 1.4 billion enrolled. Authentication failure rate ~6.5% per attempt (Policy Circle, June 2025); roughly 20.3 million failed attempts per month. 30 million ration cards cancelled due to seeding mismatches. 815 million records exposed in 2023; 29,000 biometric-cloning fraud incidents in 2024. Architecturally a centralized database lookup, not a cryptographic credential system. The canonical anti-pattern.

EU — eIDAS 2.0 / EUDI Wallet. Regulation EU 2024/1183 in force May 2024. Implementing acts December 2024. Hard deadline: 24 December 2026, every Member State must offer at least one EUDI Wallet; mandatory acceptance by relying parties December 2027. Wire protocols: OpenID4VP 1.0 Final, OpenID4VCI 1.0 Final, HAIP 1.0. Credential formats: SD-JWT VC and ISO mdoc. PQC on a separate clock: high-risk migration by 2030, full by 2035. Identity-wallet PKI explicitly flagged as longest-lead-time transition.

UK — GOV.UK One Login. 11 million users. Becomes exclusive access to central government services by end-2027. Currently lacks certification under the UK Digital Identity & Attributes Trust Framework (DIATF) due to a supplier accreditation lapse — the national IDP operating uncertified against the country's own framework.

US — mDL. No federal ID. State-level under AAMVA Implementation Guidelines v1.4–1.5. As of January 2026, 21 states plus Puerto Rico operate mDL programs accepted at TSA. ISO mdoc with COSE-signed claims. The only US system with cryptographic primitives at meaningful scale.

Canada. Verified.me effectively defunct after SecureKey’s 2022 sale. De-facto trust layer is DIACC’s Pan-Canadian Trust Framework (PCTF). No federal wallet equivalent to EUDI.

Singapore, Brazil, Nigeria. SingPass (~5M residents); gov.br (~150M users with centralizing biometric service launched 2025); Nigeria NIN with biometric General Multipurpose Card from January 2025. All architecturally centralized database-lookup; growing scale, growing exclusion.

Worldcoin / World ID. Banned, ordered to delete data, or under active investigation in Colombia, Brazil, Thailand, Philippines, Hong Kong, Spain, France, Portugal, Argentina, Kenya. Repeated regulatory finding: financial incentives (\$10–\$50 in WLD per iris scan) vitiate informed consent under GDPR-style biometric regimes. Not a deployable architecture.

9.2. Self-sovereign identity — what shipped

- W3C Verifiable Credentials Data Model 2.0 became a W3C Recommendation on 15 May 2025.
- W3C DIDs Core 1.1 reached Candidate Recommendation Snapshot on 5 March 2026; DID Resolution split into separate v0.3.
- OpenID4VP 1.0 Final and OpenID4VCI 1.0 Final published July 2025; HAIP 1.0 Final December 2025. Selected as eIDAS 2.0 ARF wire protocols.
- Hyperledger Aries archived April 2025; components migrated to the OpenWallet Foundation.
- did:web is the only DID method with material non-toy adoption (UN Transparency Protocol, Microsoft Entra Verified ID). did:ion lost momentum.

The deployed SSI surface is did:web plus OpenID4VP/VCI plus SD-JWT VC or ISO mdoc. Everything else remains research-track.

9.3. Blockchain-anchored identity

ENS retains the only material adoption (1.7M+ domains; ENSv2 mainnet February 2026). Sismo wound down late 2024. Spruce ID pivoted to government mDL integrations and now operates chain-agnostically. Proof-of-personhood remains unsolved with no production winner.

9.4. Hardware-rooted identity

- WebAuthn / Passkeys / FIDO2: more than 1 billion users with at least one passkey activated; more than 15 billion accounts support passkeys. Apple/Google/Microsoft alignment held through 2025; FIDO CTAP 2.2 advanced cross-platform portability. HID/FIDO 2025 enterprise survey reports 87% of enterprises actively deploying passkeys.
- DICE (Device Identifier Composition Engine, TCG specification): now the dominant root-of-trust pattern on constrained devices.
- Apple Secure Enclave, Google Titan-M2, Microsoft Pluton: all baseline shipping on consumer devices.
- Intel SGX: effectively abandoned for client-side use; deprecated on 11th-generation+ consumer CPUs. Server-side use shifted to Intel TDX and AMD SEV-SNP. ARM CCA Realms ships in Armv9.

9.5. Post-quantum cryptography status

- NIST FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA) published 13 August 2024. FIPS 206 (FN-DSA / Falcon) in draft.
- HQC announced March 2025 as second KEM (backup to ML-KEM).
- US OMB PQC migration guidance, required within one year of NIST standards, was overdue as of May 2026: still in draft.
- NSA CNSA 2.0: PQC required for National Security Systems by 2030; classical algorithms prohibited by 2035.
- EU joint PQC roadmap (June 2025): national strategies by end-2026; high-risk migration by 2030; full by 2035.
- **No major civic identity system runs PQC signatures in production as of May 2026.**

9.6. AI agent identity — the open lane

- Agentic AI Foundation (Linux Foundation) launched December 2025. Anthropic contributed MCP; OpenAI contributed AGENTS.md; Block contributed Goose.
- MCP authorization standardized OAuth 2.1 + PKCE in March 2025; the November 2025 release added a *server identity* primitive.
- Google A2A protocol (April 2025) uses workload-identity federation: short-lived cryptographically verifiable credentials.
- IETF Web Bot Auth draft (Cloudflare-backed) signs HTTP requests with private keys plus a published directory.

- GNAP (RFC 9635, October 2024): most plausible OAuth successor for delegated agent authorization; minimal vendor uptake.
- Model substrate attestation: *no agreed standard*. NVIDIA H100/B200, AMD SEV-SNP, Intel TDX supply hardware; the protocol layer is open.

The state of AI agent identity in 2026 is comparable to the state of SSI in 2018: many overlapping proposals, no production winner, vendors shipping ahead of standards.

9.7. Positioning fit for Paraxiom

Layer	Industry default 2026	Paraxiom choice	Lane
Wallet protocol	OID4VP/VCI + HAIP	Implement OID4VP-compatible verifier	Track
Credential format	SD-JWT VC + ISO mdoc	Support both for export	Track
DID method	did:web for interop	did:web + QH-anchored method	Track + extend
Signature crypto	RSA / ECDSA (classical)	SPHINCS+ / Falcon-512 (PQ)	Lead
Hardware root	Apple SE / Titan-M2 / Pluton	Same + DICE for edge	Track
Agent identity	MCP + OAuth 2.1 (early)	MCP + Falcon-signed + on-chain	Lead
Model attestation	None (open)	pallet_substrate_attestation	Lead
Trust framework	EU eIDAS / DIACC / state	Chain-native, bridge to civic	Sovereign

Table 4: Paraxiom positioning against the 2026 landscape. Three uncontested lanes: post-quantum civic-grade identity, agent cryptographic identity, model substrate attestation.

10. The Paraxiom architecture

We now describe the engineering realization.

10.1. Three-tier system

The Paraxiom identity stack is decomposed into three layers of storage with three distinct trust and performance profiles:

Layer	Holds	Speed	Mutability
QuantumHarmony chain	Identity records, hashes, CIDs, commitments, signatures	Slow	Append-only
IPFS	Encrypted payloads, KYC bundles, vault ciphertext	Medium	Immutable (b
Postgres + Redis	Denormalized indexes, query caches	Fast	Rebuildable n

Table 5: The three storage layers of the Paraxiom identity stack.

The chain is the source of truth for cryptographic claims. IPFS is the canonical bulk-payload store, content-addressed by CID. The Postgres+Redis mirror is rebuildable from chain plus IPFS without information loss (modulo unrecoverable plaintexts, which are protected by deletion of the master key per Theorem 10.1).

10.2. Identity strata

The five strata of Theorem 6.1 are realized on chain through the existing identity-registry pallet (extended) and a small number of companion pallets:

```
pub struct Identity<AccountId, BlockNumber> {
    pub id: u64,
    pub owner: AccountId,
    pub public_key: BoundedVec<u8, ConstU32<2048>>,
    pub role: IdentityRole,
    pub status: IdentityStatus,
    pub reputation_bps: u32,
    pub created_at_block: BlockNumber,
    pub created_by_tx_hash: H256,
    pub key_rotation_history: BoundedVec<KeyRotation, ConstU32<32>>,

    /// Stratum tag (S1, S2, S4, S5).
    pub stratum: Stratum,

    /// Parent identity for S2 and S4 (required by Theorem 6.2).
    pub parent_identity: Option<u64>,

    /// Tenant context for S2.
    pub tenant_id: Option<BoundedVec<u8, ConstU32<64>>>,

    /// Custody mode (SelfSovereign / NcCustodial / TenantCustodial).
    pub custody: CustodyMode,

    /// KYC bundle hash for S1 (Falcon-signed by KYC issuer).
    pub kyc_bundle_hash: Option<[u8; 32]>,

    /// Instance anchor for S4 agents (per Corollary 5.1).
    pub instance_anchor: Option<[u8; 32]>,
}
```

10.3. Companion pallets

pallet-onboarding-commit (extended): a generalized commit-reveal flow with bond and slashable dispute window. Used to bootstrap new S1 identities, agent S4 identities, and validators.

axiom-attestation (extended): on-chain Falcon-signed attestations with IPFS CIDs for full payloads. Extended with a VerifierKind enum to dispatch question-generation, PR-code-review, vault-commitment, persona-attestation, and existing Axiom-task variants.

pallet-credential-vault (new): on-chain commitments to off-chain envelope-encrypted credentials. Stores (ciphertext_cid, ciphertext_hash, iv_hint, service, kind, created_at, revoked_at).

pallet-reputation-tokens (new): soulbound earned reputation tokens with minting policies indexed by behavior. Distinct from the governance-adjustable reputation_bps of the identity registry.

pallet-substrate-attestation (new): per Section 7, publishes LLM weights provenance on chain.

10.4. Credential vault flow

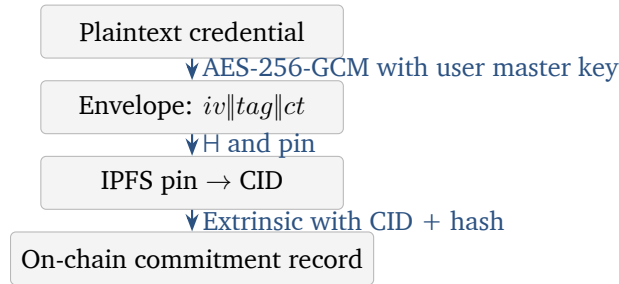


Figure 2: Credential vault encryption flow. Plaintext never leaves the client (v0.2+) or server boundary (v0.1); IPFS holds ciphertext only; chain holds commitment.

10.5. Crypto-shredding completeness

A key correctness property of the credential vault: revocation through master-key deletion must render all encrypted credentials inaccessible, *even if the ciphertext remains pinned to IPFS*.

Theorem 10.1 (Crypto-shredding completeness). *Let k be a master key over a strongly-secure AEAD scheme (AES-256-GCM under the standard model). Let $\{C_1, \dots, C_n\}$ be a set of ciphertexts $C_i = \text{Enc}_k(m_i, iv_i)$ for messages m_i and unique nonces iv_i . Suppose k is securely erased from all storage controlled by the holder. Then, for any computationally-bounded adversary $\mathcal{A}$ with access to all C_i but not to k ,*

$$\Pr[\mathcal{A}(C_1, \dots, C_n) = m_i] \leq \frac{1}{|\mathcal{M}|} + \text{negl}(\lambda),$$

where $\mathcal{M}$ is the plaintext domain and λ is the AEAD security parameter.

Proof. By the IND-CPA security of AES-256-GCM under the standard model. $\text{negl}(\lambda)$ is the adversary's IND-CPA advantage at security parameter $\lambda = 256$. Crucially, AES-256-GCM's security guarantee binds the security to key secrecy, not to ciphertext exclusivity: public ciphertext does not weaken the bound. Erasing k thus removes all information-theoretic distinguishability from the chosen-message oracle. ■

Corollary 10.2 (CCPA crypto-shredding). *Compliance with right-to-erasure obligations (CCPA §1798.105, GDPR Art. 17) can be discharged by deleting the user’s master key from all holder-controlled stores, without requiring unpinning of IPFS-resident ciphertexts.*

10.6. Sync mechanism

The Postgres+Redis mirror in the gov-app subscribes to chain events via Substrate’s WebSocket RPC. Events from identity-registry, axiom-attestation, pallet-reputation-tokens, and pallet-credential-vault are decoded and applied to the relational schema; Redis caches are invalidated on the keyed indexes.

10.7. Verifier at action time

Theorem 10.3 (Verification-time accountability). *Let $\mathcal{I}$ be an identity attempting an action α at time t . The `verify_chain( $\mathcal{I}, \alpha, t$ )` helper returns $\top$ iff (i) $\mathcal{I}$ has an active L_3 key at t , (ii) $\mathcal{I}$ ’s $\prec$ -ancestry terminates in finite steps at an S1 with a non-empty L_4 claim or in a stake-bonded entity, and (iii) $\mathcal{I}$ ’s L_5 reputation satisfies the action’s minimum threshold θ_α .*

Proof. Conditions (i) and (iii) are checked by direct query against identity-registry and pallet-reputation-tokens. Condition (ii) terminates in finite steps by Theorem 6.2. ■

11. Future scenarios

We project three scenarios for the trajectory of identity past 2030.

11.1. Dissolution

Identity collapses if (a) AI agents proliferate faster than accountability chains can be maintained; (b) CRQC arrives before PQC migration completes for civic-identity infrastructure; (c) states lose capacity to issue civic identity; (d) privacy norms preclude reputation tracking. Non-zero probability; the canonical reason this work is timely.

11.2. Fragmentation

Identity persists but jurisdictionalizes: each major bloc operates its own infrastructure with limited interoperability. This is the trajectory the world is presently on. Paraxiom’s chain-native, state-agnostic substrate functions as a bridge across fragmented zones, attesting to jurisdictional claims when needed via `kyc_bundle_hash` but not depending on any single jurisdiction’s continued existence.

11.3. Augmentation

Identity extends: humans, agents, swarms, organizations, protocols all carry first-class cryptographic identity with explicit accountability chains and portable reputation. The five-stratum design of Theorem 6.1 is a first draft of an augmented identity ontology; further strata (swarm, protocol, hybrid) extend the schema.

The outcome among these three depends not on inevitability but on what is built. Dissolution occurs by default if no one builds the infrastructure; fragmentation if states build it without interoperability; augmentation if deliberate choice produces post-quantum, decentralized, accountability-preserving substrates.

12. Implementation roadmap

We sequence the engineering work.

1. **Identity-registry extensions** (~3 days). Non-breaking; ships first. Adds stratum, parent_identity, tenant_id, custody, kyc_bundle_hash, instance_anchor fields.
2. **pallet-onboarding-commit generalization** (~2 days). Per-role bonds; supports S1/S4/validator commit-reveal.
3. **axiom-attestation VerifierKind extension** (~2 days). Adds QuestionGen, PrCodeReview, VaultCommitment, PersonaAttestation variants alongside the existing AxiomTask.
4. **pallet-credential-vault** (~3 days). On-chain commitments to IPFS-resident encrypted credentials. Realizes Theorem 10.1.
5. **pallet-reputation-tokens** (~4 days). Earned, soulbound, threshold-minted tokens.
6. **pallet-substrate-attestation** (~3 days). Per Section 7, publishes LLM substrate provenance.
7. **Gov-app event subscriber** (~5 days). Subxt or Polkadot.js client; consumes chain events; populates Postgres mirror.
8. **Spec version bump + on-chain migration** (~3 days). Migrates existing identities to the extended schema.

Total estimated effort: 3–4 weeks of focused work, parallelizable with gov-app v0.2 work that depends on the migrated schema.

13. Conclusion

Identity is engineering. The accountability chain does not extend itself; the post-quantum signatures do not sign themselves; the strata do not enumerate themselves. The substrates we now share with machines and AI agents will, by the early 2030s, demand identity infrastructure that survives a quantum-cryptographic transition, that scales without requiring social acquaintance, that holds open the possibility of new kinds of accountability loci we cannot fully enumerate today.

We have stated identity formally as a layered accountability chain above a founding individuation event, proven four invariants (founding uniqueness, tier monotonicity, cryptographic survival, crypto-shredding completeness), surveyed the 2026 deployed landscape and located Paraxiom’s three uncontested lanes (PQ civic-grade, agent cryptographic identity, model substrate attestation), and specified the on-chain plus IPFS plus mirror architecture that realizes the model.

The question with which this treatise was commissioned — “will we have identity in the future?” — admits a precise answer. Yes, if built. The pallets specified in Section 10 are the construction.

A. Detailed proofs

A.1. Founding uniqueness (full)

Full proof of Theorem 2.8. We argue from Theorem 2.3, Theorem 2.1, and Theorem 2.5.

Let $\mathcal{I} = (\mathcal{S}, \mathcal{F}, \mathcal{A}, \mathcal{R})$ be an identity locus and suppose for contradiction that there exist two founding-event candidates $\mathcal{F}_1 = (s, t_0^{(1)}, \pi_1)$ and $\mathcal{F}_2 = (s, t_0^{(2)}, \pi_2)$, both consistent with $\mathcal{I}$ in the sense that every claim in $\mathcal{A}$ post-dates the candidate’s timestamp.

Case 1: $t_0^{(1)} < t_0^{(2)}$. Then there exist points in time $t \in (t_0^{(1)}, t_0^{(2)})$ at which $\mathcal{S}$ was already individuated relative to $\mathcal{F}_1$ but, by hypothesis, not yet individuated relative to $\mathcal{F}_2$. Any claim made in this interval — and Theorem 2.5 permits such claims to exist in $\mathcal{A}$ relative to $\mathcal{F}_1$ — would violate Theorem 2.5 relative to $\mathcal{F}_2$. Either $\mathcal{F}_2$ does not consistently anchor $\mathcal{I}$, or no claims exist in the interval (in which case $t_0^{(2)}$ can be taken as $t_0^{(1)}$ without loss of information). In both sub-cases we collapse to a single founding event.

Case 2: $t_0^{(1)} = t_0^{(2)}$. Then both events name the same moment of individuation of the same substrate; up to the witness set, they coincide. The witness sets may be merged: $\pi = \pi_1 \cup \pi_2$ produces the most informative single $\mathcal{F}$, and the two-event presentation reduces to a single-event presentation.

In both cases the two candidates collapse to one, contradicting the assumption of distinctness. ■

A.2. Stratum well-foundedness (full)

Full proof of Theorem 6.2. We rely on three facts about the chain registration protocol:

1. Every S2 or S4 registration extrinsic requires a non-null `parent_identity` field; the chain validates that the named parent exists prior to applying the registration.
2. S5 registration requires either a multi-signature attestation of S1 founders or a jurisdictional L_4 claim. Both terminate finitely at S1 or at a jurisdictional authority treated as an S1 surrogate.
3. Every identity in `identity-registry` carries a `created_at_block` field, strictly less than the block in which any of its children are registered.

Suppose for contradiction an infinite descending chain $\mathcal{I}_0 \succ \mathcal{I}_1 \succ \mathcal{I}_2 \succ \dots$. By fact 3, $\text{block}(\mathcal{I}_0) > \text{block}(\mathcal{I}_1) > \text{block}(\mathcal{I}_2) > \dots$, an infinite strictly decreasing sequence in $\mathbb{N}$, which is impossible.

Thus every descending chain terminates in finite steps. By facts 1 and 2, the terminus is either an S1 or an S1 surrogate. In either case, finite-step termination at an S1-equivalent is guaranteed. ■

A.3. Cryptographic survival (full)

Full proof of Theorem 8.1. Forward direction. For $\Sigma \in \Sigma_{\text{pQ}}$ each scheme has been demonstrated EUF-CMA-secure against any polynomial-time quantum adversary under the standard model, per the security reductions underlying NIST FIPS 203–205. Let $\mathcal{Q}$ be such a quantum adversary with running time bounded by $\text{poly}(\lambda)$. The EUF-CMA security definition states:

$$\Pr[\text{EUF-CMA}_{\mathcal{Q}}(\Sigma, \lambda) = 1] \leq \text{negl}(\lambda).$$

This implies that the probability of any forgery against any single claim a_i in $\mathcal{A}$ is at most $\text{negl}(\lambda)$. By a union bound over the n claims, the probability of any forgery anywhere in $\mathcal{A}$ is at most $n \cdot \text{negl}(\lambda)$, which remains negligible for any polynomial-size $\mathcal{A}$.

Reverse direction. For $\Sigma' \in \Sigma_{\text{cl}}$, Shor’s algorithm [4] solves the discrete logarithm problem in $\mathbb{Z}_p^*$ and on elliptic curves over $\mathbb{F}_p$ in time polynomial in $\log p$. Given a CRQC of size sufficient to implement Shor at the relevant key size, the adversary can recover sk from pk with probability $1 - \text{negl}'(\lambda')$, after which forgery is trivial via Sign_{sk} . The probability bound on forgery is hence at least $1 - \text{negl}'(\lambda')$, where $\text{negl}'(\lambda')$ is characterized by the CRQC’s residual error, dominated by the classical key length λ' . ■

A.4. Crypto-shredding completeness (full)

Full proof of Theorem 10.1. We reduce to the IND-CPA security of AES-256-GCM. Suppose for contradiction that $\mathcal{A}$ achieves

$$\Pr[\mathcal{A}(C_1, \dots, C_n) = m_i] > \frac{1}{|\mathcal{M}|} + \epsilon$$

for non-negligible ϵ , for some target index i . Construct an IND-CPA adversary $\mathcal{B}$ against AES-256-GCM as follows: $\mathcal{B}$ receives an IND-CPA challenge oracle, queries it on all $\{m_1, \dots, m_n\}$ to obtain ciphertexts identically distributed to those $\mathcal{A}$ would receive, then runs $\mathcal{A}$ on these ciphertexts. The output of $\mathcal{B}$ is $\mathcal{A}$'s output.

If $\mathcal{A}$ recovers m_i with advantage ϵ over the $1/|\mathcal{M}|$ baseline, then $\mathcal{B}$ distinguishes the IND-CPA challenge with the same advantage. By IND-CPA security of AES-256-GCM, this advantage is bounded by $\text{negl}(\lambda)$ where $\lambda = 256$ is the AES key length. Hence $\epsilon \leq \text{negl}(\lambda)$, contradicting the supposed non-negligibility. ■

B. Pallet API specifications

B.1. identity-registry (extended)

```
pub enum Stratum { S1Person, S2Scoped, S4AgentInstance, S5CrossTenant }

pub enum CustodyMode { SelfSovereign, NcCustodial, TenantCustodial }

pub enum IdentityRole {
    Shield, Agent, OracleReporter, Validator,
    Citizen, Decideur, Fournisseur, Service,
}

pub enum IdentityStatus {
    Active, Suspended, Revoked, Deceased, Decommissioned, Dissolved
}

/// Register an S1 identity, optionally with KYC bundle.
pub fn register_identity_s1(
    origin: OriginFor<T>,
    public_key: BoundedVec<u8, ConstU32<2048>>,
    kyc_bundle_hash: Option<[u8; 32]>,
    custody: CustodyMode,
) -> DispatchResult;

/// Delegate to a scoped S2 identity from a caller S1.
pub fn delegate_to_scope(
    origin: OriginFor<T>,
    tenant_id: BoundedVec<u8, ConstU32<64>>,
    role: IdentityRole,
) -> DispatchResult;

/// Register an S4 agent instance under a parent identity.
```

```

pub fn register_agent_instance(
    origin: OriginFor<T>,
    parent_identity: u64,
    agent_kind: BoundedVec<u8, ConstU32<64>>,
    public_key: BoundedVec<u8, ConstU32<2048>>,
    instance_anchor: [u8; 32],
) -> DispatchResult;

/// Verify the accountability chain for an action.
pub fn verify_chain(
    identity_id: u64,
    action_kind: ActionKind,
    block: BlockNumberFor<T>,
) -> Result<ChainVerdict, DispatchError>;

```

B.2. pallet-credential-vault

```

pub struct Commitment<BlockNumber> {
    pub holder_identity: u64,
    pub service_name: BoundedVec<u8, ConstU32<64>>,
    pub credential_kind: CredentialKind,
    pub ciphertext_cid: BoundedVec<u8, ConstU32<128>>,
    pub ciphertext_hash: [u8; 32],
    pub iv_hint: [u8; 12],
    pub created_at: BlockNumber,
    pub revoked_at: Option<BlockNumber>,
}

pub fn commit(
    origin: OriginFor<T>,
    service_name: BoundedVec<u8, ConstU32<64>>,
    credential_kind: CredentialKind,
    ciphertext_cid: BoundedVec<u8, ConstU32<128>>,
    ciphertext_hash: [u8; 32],
    iv_hint: [u8; 12],
) -> DispatchResult;

pub fn revoke(
    origin: OriginFor<T>,
    commitment_id: u64,
) -> DispatchResult;

pub fn attest_use(
    origin: OriginFor<T>,
    commitment_id: u64,
    caller_context: BoundedVec<u8, ConstU32<128>>,
) -> DispatchResult;

```

B.3. pallet-substrate-attestation

```

pub struct SubstrateAttestation<BlockNumber> {
    pub weights_hash: H256,
    pub training_run_merkle: H256,
    pub arch_spec_hash: H256,
    pub publisher: u64, // identity-registry ID
    pub publisher_sig: BoundedVec<u8, ConstU32<2048>>,
    pub ipfs_cid_card: BoundedVec<u8, ConstU32<128>>,
    pub published_at: BlockNumber,
    pub revoked_at: Option<BlockNumber>,
}

pub fn attest_substrate(
    origin: OriginFor<T>,
    weights_hash: H256,
    training_run_merkle: H256,
    arch_spec_hash: H256,
    publisher_falcon_sig: BoundedVec<u8, ConstU32<2048>>,
    ipfs_cid_card: BoundedVec<u8, ConstU32<128>>,
) -> DispatchResult;

```

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Document version: 1.0, May 15, 2026. *Companion materials:* IDENTITY-ARCHITECTURE.md (narrative version); IDENTITY-PALLETS-DESIGN.md (engineering specification); GOVERNANCE-ROADMAP.md §10 (product roadmap). *Repository:* <https://github.com/Paraxiom/quantumharmony> and companion documents in the NeoCarbone lovable-bridge repository.